Structural Equation Models

Advanced Master in Agricultural Economics and Policy

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Attitudinal Scales for the Study
of Consumer Preferences
April 30 and May 7, 2026

Roadmap

20/04 and 23/04	Hello, Psychometrics!	✓
24/04	Reliability and Validity of a Measure	✓
24/04 and 27/04	Latent Variables: Reflective or Formative?	✓
30/04 and 07/05	Structural Equation Models	◯

Recommended references:

- Chapters 4 and 7 from Jhangiani et al.2019)
- Chapters 7, 10 and 14 from Olivero and Russo2022)
- Chapter 12 from Mehmetoglu and Jakobsen2022)

Jhangiani, R. S., Chiang, I. A., Cuttler, C., and Leighton, D. C. (2019). *Research Methods in Psychology*. Kwantlen Polytechnic University, 4th edition.

Mehmetoglu, M. and Jakobsen, T. G. (2022). *Applied Statistics Using Stata: A Guide for the Social Sciences*. SAGE Publications Ltd, 2nd edition.

Olivero, N. and Russo, V. (2022). *Psicologia dei Consumi*. McGraw-Hill Education, 3rd edition.

Exam

Each of you has been assigned a scientific article. The objective of the exam is to critically analyze the use of psychometric tools within the study and propose a motivated extension, based on your theoretical and methodological knowledge. The report must be submitted by **27/06/2025**.

Info and material:
`https://data.califano.xyz/esame`

Before starting...

https://www.stata.com/manuals/sem.pdf

Before starting...

Remarks and examples

Structural equation modeling encompasses a broad array of models from linear regression to measurement models to simultaneous equations, including along the way confirmatory factor analysis (CFA), correlated uniqueness models, latent growth models, multiple indicators and multiple causes (MIMIC) models, and item response theory (IRT) models.

Structural equation modeling is not just an estimation method for a particular model in the way that Stata's `regress` and `probit` commands are, or even in the way that `stcox` and `mixed` are. Structural equation modeling is a way of thinking, a way of writing, and a way of estimating.

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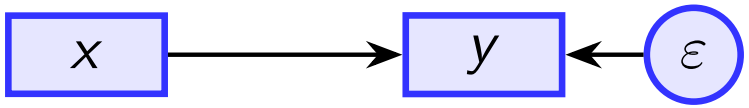

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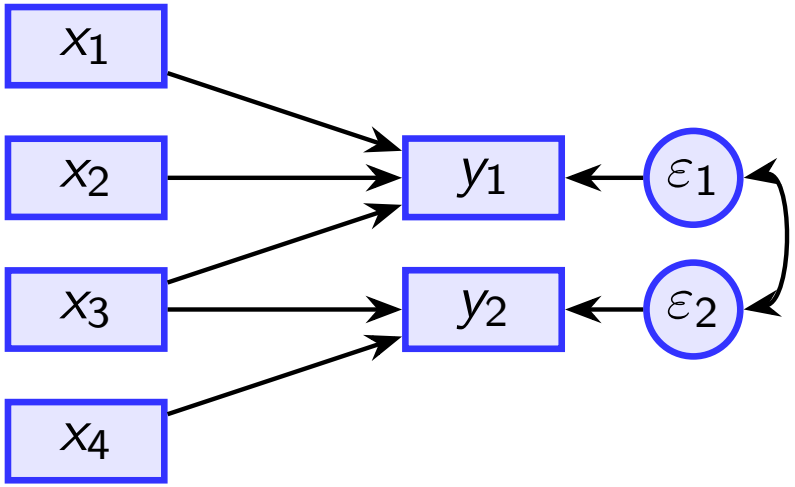


Moving on to...

A bit of terminology

A bit of history

A bit of theory

A bit of Stata

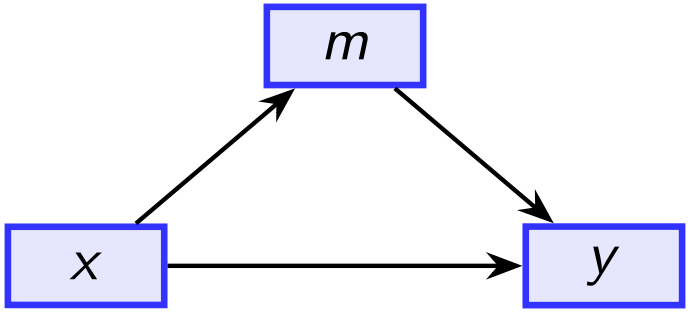
Types of SEM

- Structural equation modeling allows for estimating the relationships between observed and/or latent variables by studying the variance and covariance matrix. For this reason, it is also called **Covariance Based (CB) SEM**[1].
- Since it is used to estimate multiple equations simultaneously, the models are described and reported in the form of block diagrams.
- As noted by the Stata manual, many different models fall under the umbrella of SEM...

[1] Also to distinguish it from Partial Least Squares (PLS) SEM.

A few examples

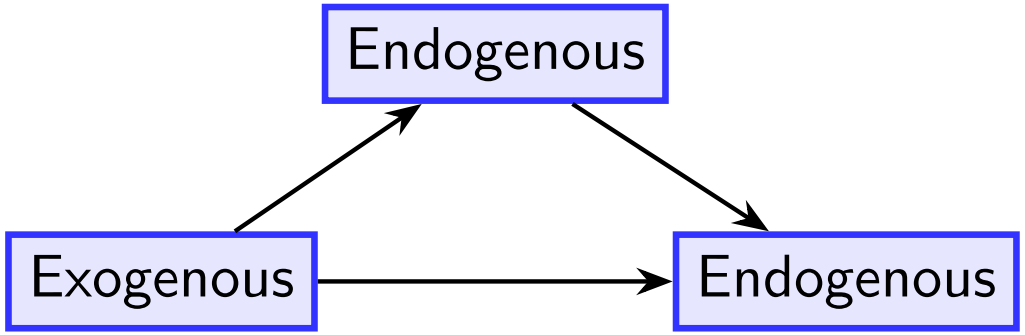
Path Analysis (PA)



Relationships between manifest variables are estimated.

A few examples

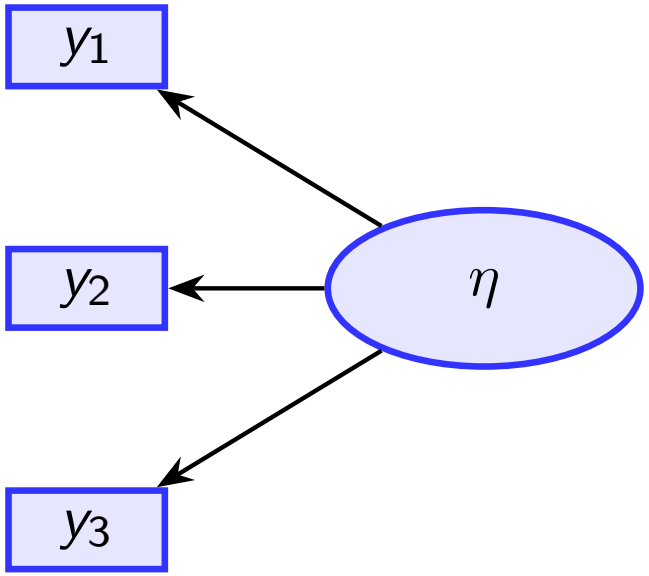
Path Analysis (PA)



In these models it is complicated to speak of dependent and independent variables. Variables that are not explained in the model are defined as exogenous, while those that are explained are defined as endogenous.

A few examples

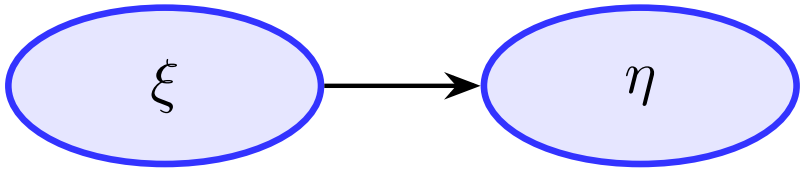
Confirmatory Factor Analysis (CFA)



Relationships between manifest and latent variables are estimated.

A few examples

Latent Path Analysis (LPA)

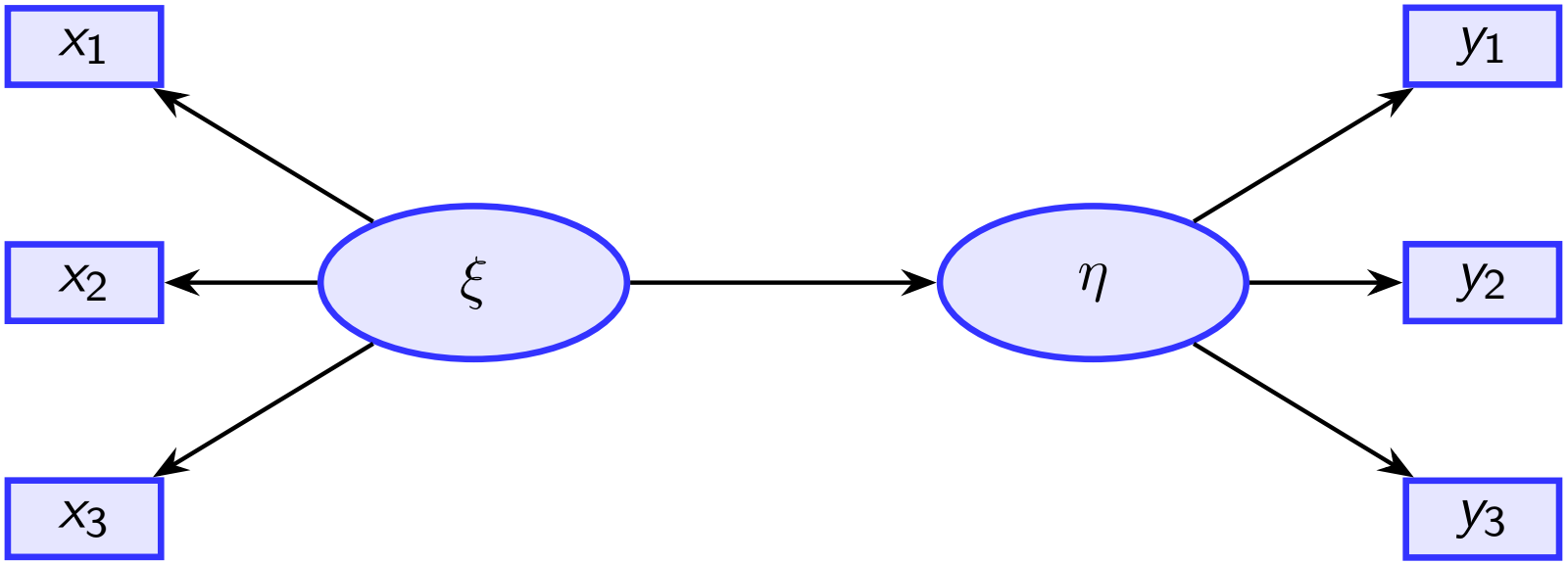


Relationships between latent variables are estimated.

A few examples

CFA + LPA

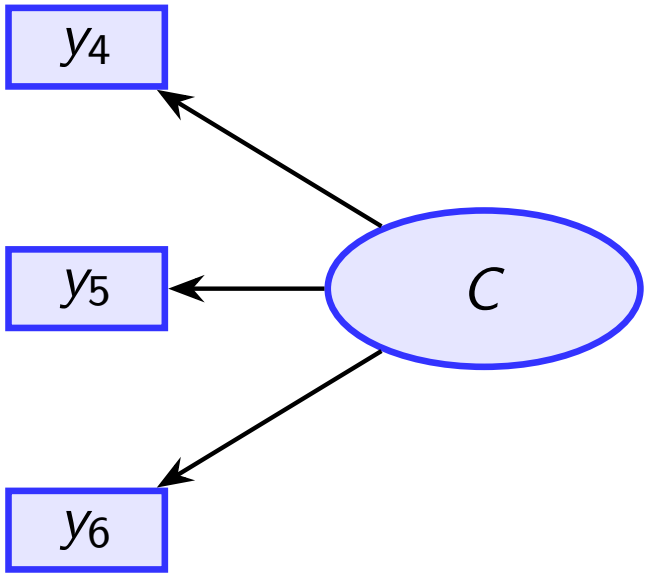
What is commonly meant by SEM



In these cases the CFA part is defined as the **measurement model** *or* **outer model**. *The LPA (or PA) part, on the other hand, as the* **structural model** *or* **inner model**.

A few examples

Are latent variables only continuous?



No. We can hypothesize the existence of unobservable (or unobserved) classes and estimate them from observed indicators. Latent categorical variable SEM includes Latent Class Analysis, Latent Profile Analysis, and other finite mixture models.

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A brief history of SEM

Origins

SEM analysis was born from the integration of:

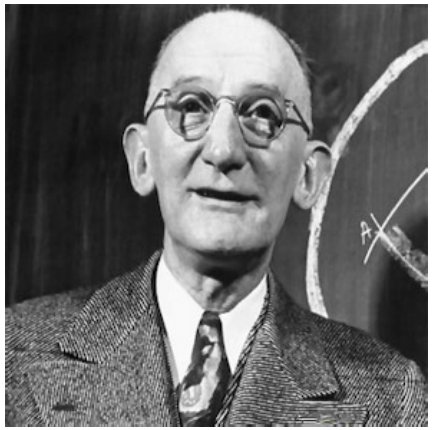
- **Path Analysis** (Sewall Wright, 1921)
- **Factor Analysis** (Spearman, 1904; Thurstone, 1930)
- **Latent variable models** (Jöreskog, 1969)

SEM unifies causal and measurement approaches into a single statistical structure.

Sewall Wright

Charles Spearman

Louis Thurstone



Karl Jöreskog

A brief history of SEM

CB-SEM: Covariance-Based SEM

- Formalized by Karl Jöreskog (1969) with LISREL;
- Goal: reproduce the observed **covariance matrix**;
- Based on **maximum likelihood**.

PLS-SEM: Partial Least Squares SEM

- Introduced by Herman Wold (1970s);
- **Component-based**, iterative approach;
- Focused on **maximizing explained variance**;
- Preferable in case of: small samples, formative models, non-normality.

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Analysis process

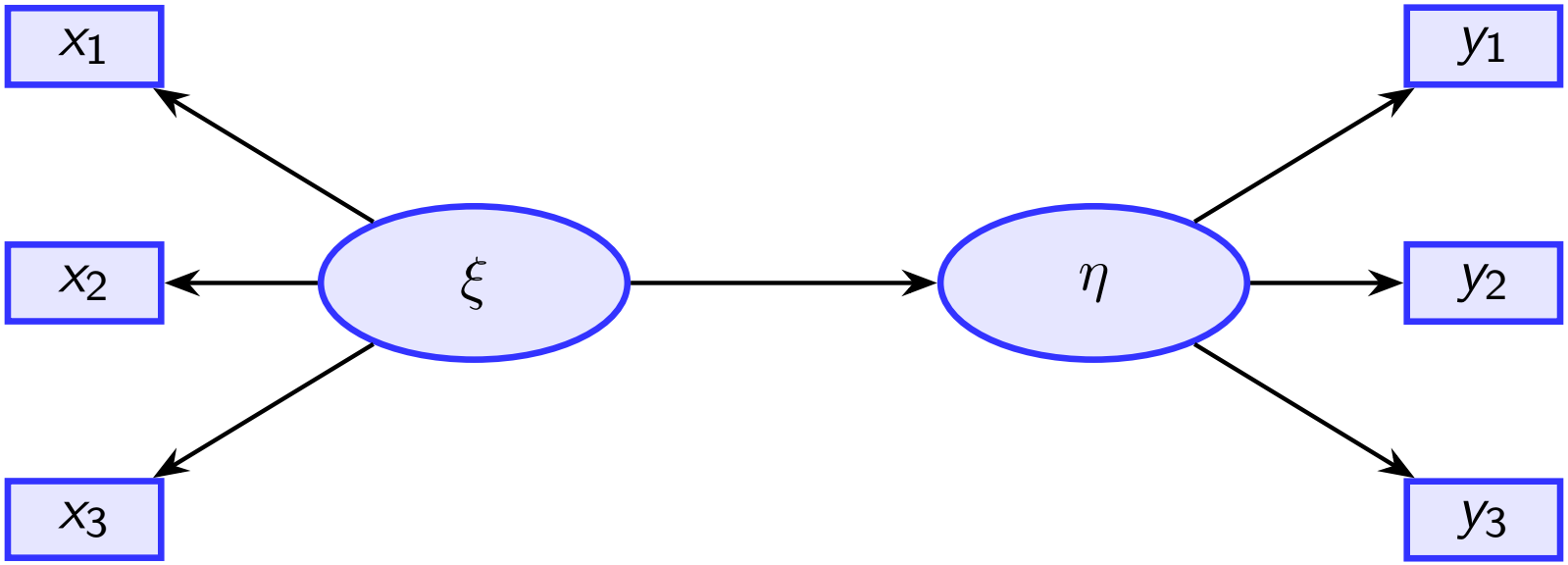
SEM analysis goes through four phases, the same for **CB-SEM** and **PLS-SEM**. However, the specific criteria to be used differ, as these models differ in functioning and purposes of use (e.g., Reflective or Formative LV). We will focus on the four phases of the process with a focus on **CB-SEM**:

1. Model specification
2. Model identification
3. Model estimation
4. Model evaluation

Model specification

Phase 1:

The first phase consists of defining the theoretical model and the hypothesized relationships. In many software programs (including Stata), model specification can occur either through drawing diagrams or through writing code.




```
. sem (Xi -> x1 x2 x3) (Eta -> y1 y2 y3) (Xi -> Eta)
```

Model identification

Phase 2:

Unfortunately, not everything that can be drawn can also be estimated. The **T-Rule**, for example, is a necessary (but not sufficient) condition for a model to be identifiable:

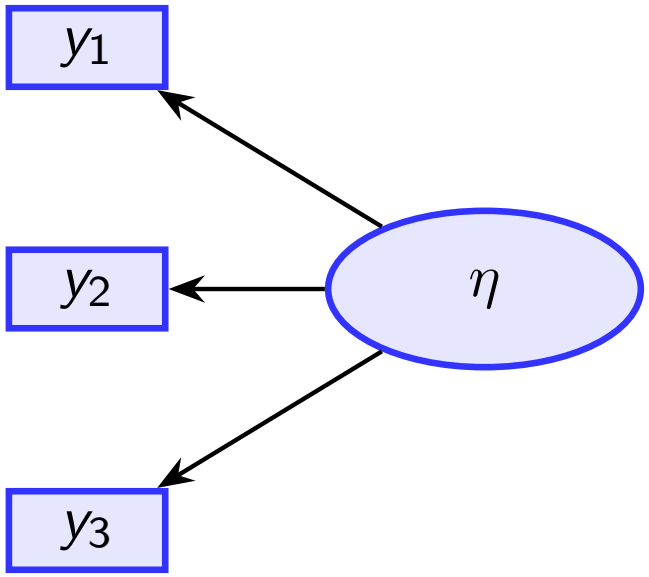
$$t \leq \frac{p(p+1)}{2}$$

- t is the number of parameters to be estimated (arrows in the diagrams);
- p is the number of manifest variables in the model.

- If $t < \frac{p(p+1)}{2}$ the model is *over-identified*: estimable and testable;
- If $t = \frac{p(p+1)}{2}$ the model is *just-identified*: estimable but not testable (degrees of freedom $= 0$);
- If $t > \frac{p(p+1)}{2}$ the model is *under-identified*: cannot be uniquely estimated.

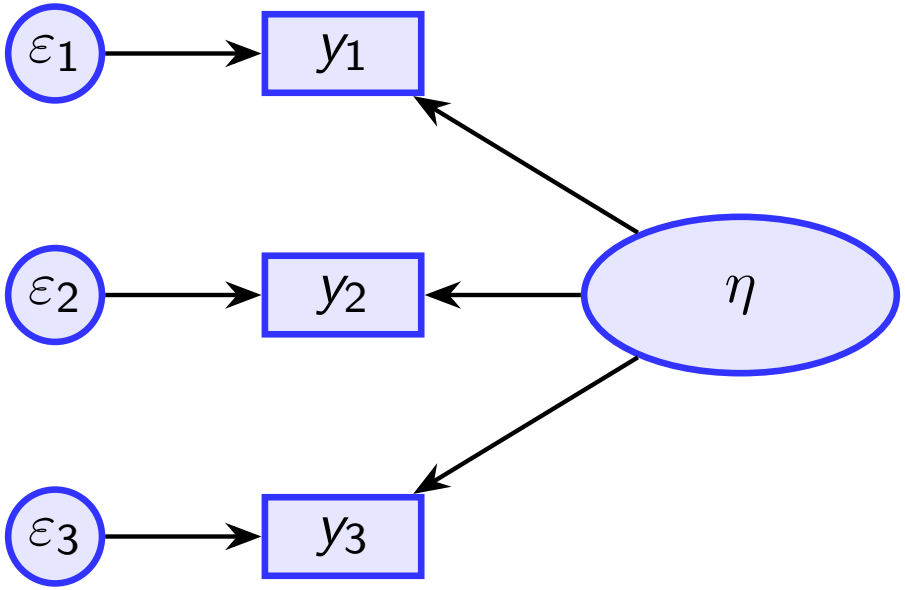
Model identification

$$t \leq \frac{p(p+1)}{2} \qquad ?$$



Model identification

$$t \leq \frac{p(p+1)}{2} \qquad ?$$



Model estimation

Phase 3:

Model estimation consists of calculating the parameter values that best reproduce the observed covariance matrix (typically through *maximum likelihood*).

Model evaluation

Phase 4:

Model evaluation occurs at three levels:

1. **Global goodness-of-fit**:
   - χ^2 test (better non-significant)
   - RMSEA < 0.06
   - CFI, TLI > 0.95
   - SRMR < 0.08
2. **Measurement model**:
   - λ_i (factor loadings) > 0.5 and significant
   - CR (Composite Reliability) > 0.7
   - AVE (Average Variance Extracted) > 0.5
   - Discriminant validity (e.g., Fornell-Larcker criterion)
3. **Structural model**:
   - β (path coeff.) significant
   - R^2 acceptable for endogenes
   - Significant direct, indirect and total effects (if relevant)

Modification Indices

What are they?

Modification Indices (MI) signal potential improvements to the model by introducing unspecified parameters (e.g., residual covariances).

Warning: suggested changes should be accepted only if theoretically justified, to avoid *overfitting*.

By introducing new parameters to estimate, we restart from Phase 1!

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clear all ;)

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